MUSANZE HARVEST INTELLIGENCE SERVICE: TECHNICAL REPORT

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1. SCENARIO AND STAKEHOLDER ECOSYSTEM

The Irish potato (*Solanum tuberosum*) sub-sector is the economic backbone of Musanze District and the wider volcanic highlands of Northern Province, Rwanda. Cooperative unions such as regional potato farmer collectives receive thousands of seasonal farm production records from smallholder members across high-elevation parcels. Historically, these field records arrive in fragmented, heterogeneous spreadsheets plagued by missing measurements, transcription typos, and subjective yield expectations. Field extension officers and cooperative managers urgently require an evidence-based, automated intelligence tool that operates reliably offline in rural field stations, provides validated crop predictions, and integrates with prospective mobile applications.

2. OPERATIONAL DECISION PROBLEMS & ANALYTICAL QUESTIONS

This decision-support prototype translates real cooperative operational bottlenecks into three machine learning tasks:

- **Task 1 (Regression – Harvest Yield Estimation):** How many metric tons of Irish potatoes will a given parcel produce based on climatic conditions (rainfall, temperature), volcanic soil moisture, chemical/organic fertilizer inputs, parcel surface area, cropping season (A vs B), and cooperative support?
- **Task 2 (Classification – High-Risk Vulnerability Triage):** Which smallholder farms are in jeopardy of "high production risk" — operationalized rigorously as potato yield per hectare falling strictly below the historical median across regional holdings?
- **Task 3 (Clustering – Agronomic Farm Typology):** How can cooperative farm parcels be segmented into homogeneous operational clusters based strictly on physical inputs (without target leakage) to guide tailored agronomic training, seed distribution, and fertilizer subsidies?

3. ETHICAL BOUNDARIES, OPERATIONAL LIMITS & HUMAN OVERSIGHT

The service is engineered as a classroom decision-support prototype. To guarantee responsible deployment and prevent algorithmic harm to vulnerable agrarian communities, the following hard constraints are established:

> **Mandatory Operational & Ethical Guardrails:**
> • **Advisory Nature:** The intelligence service functions strictly as an advisory aid; it must never claim to replace in-person certified agronomists or laboratory soil chemical analyses.
> • **Welfare & Financial Protection:** System risk predictions are prohibited from being used by financial institutions, insurance underwriters, or cooperative boards to deny access to credit, crop insurance payouts, land tenure certifications, or national subsidy programs.
> • **Human-in-the-Loop Requirement:** All high-risk automated alerts mandate on-site physical verification by a cooperative field officer before triggering corrective agricultural interventions.

DATASET ARCHITECTURE AND CRYPTOGRAPHIC REPRODUCIBILITY

1. FEATURE DATA DICTIONARY AND GENERATION RULES

Attribute	Type	Unit / Range	Generation Distribution	Functional Role
`rainfall_mm`	Float	[600.0, 1400.0] mm	Uniform(600, 1400)	Seasonal rainfall input feature
`avg_temp_c`	Float	[12.0, 26.0] °C	Uniform(12, 26)	Mean ambient temperature input feature
`soil_moisture_pct`	Float	[20.0, 80.0] %	Uniform(20, 80)	Volumetric soil water content feature
`fertilizer_kg`	Float	[0.0, 250.0] kg	Uniform(0, 250)	Total NPK / organic fertilizer applied
`field_size_ha`	Float	[0.2, 4.0] ha	Uniform(0.2, 4.0)	Cultivated potato parcel area
`season`	Category	'A' (Main), 'B' (Short)	Discrete Uniform {A: 0.5, B: 0.5}	Cropping season (+0.50t A, -0.20t B)
`cooperative_support`	Binary	0 (No), 1 (Yes)	Bernoulli (p = 0.5)	Access to cooperative inputs (+0.70t)
`yield_tons`	Float	≥ 0.50 tons	Analytical Formula + Normal Noise	Continuous Regression Target
`high_risk`	Binary	0 (Low), 1 (High)	Yield/ha < Regional Median	Binary Classification Target

2. INDIVIDUAL SEED RULE & MATHEMATICAL OBSERVATION COUNT

The data synthesis algorithm is locked to the last 4 digits of the student's registration number (`25/28110 --> LLLL = 8110`). In strict accordance with the individual data signature specifications:

```
Random Number Generator: numpy.random.default_rng(8110) Base Row Count: 240 + (LLLL mod 41) = 240 + (8110 mod 41) = 240 + 33 = 273 base rows Missing Count m: round(0.05 * 273) = round(13.65) = 14 unique row positions per faulted feature Appended Duplicate Rows: Exactly 3 identical duplicate rows appended to the end of the base table Total Serialized CSV Records: 273 + 3 = 276 total records
```

3. DELIBERATE DATA QUALITY DEFECTS & CLEANSING EVIDENCE

Real-world agricultural spreadsheets contain errors. The generator deliberately created two major defects:

- **Deliberate Missing Values:** 14 unique observations in `soil_moisture_pct` and 14 distinct unique observations in `fertilizer_kg` were randomly set to null (NaN).
- **Deliberate Redundant Duplicates:** Three exact duplicate records were appended to the base records, expanding the CSV to 276 rows.
- **Automated Pipeline Cleansing:** The ingestion validator (`src/validation.py`) detected and removed the 3 duplicate rows without human intervention, restoring the true population of 273 unique observations. Missing values were subsequently processed via train-only median imputation.

4. AUTHORITATIVE SHA-256 DATASET FINGERPRINT

The dataset was formatted with `float_format="%.4f"` and hashed using SHA-256. Running `run_all.py` on any clean operating system produces this exact verifiable fingerprint:

```
6f94df679bcd2c644aab4eceed2cfb02b09a5ddddbfa920c7dc2c018d59e1d5d
```

PIPELINE ARCHITECTURE & ANTI-LEAKAGE CONTROLS

1. TRAIN / TEST SEPARATION PROTOCOL

The cleansed population of 273 unique farm records was partitioned into an 80% training set (218 records) and a 20% held-out test set (55 records) using seed `8110`. The test set was immediately locked and sequestered. No test records were accessible during feature transformer fitting, model parameter optimization, or cluster center calculation.

2. TRAIN-ONLY PREPROCESSING FLOW

To eliminate data snooping, all data transformations are encapsulated in `PreprocessingPipeline`, which fits transformation parameters strictly on the training partition:

- **Median Imputation:** Missing numerical values in `soil_moisture_pct` and `fertilizer_kg` are imputed using training medians (`soil_moisture_median = 49.60%`, `fertilizer_median = 127.35 kg`).
- **Standardization (Z-Score Scaling):** Numerical features are standardized using training sample means (K\mu_{\text{train}}K) and standard deviations (K\sigma_{\text{train}}K). Test features and live API inference inputs are transformed using these stored parameters.
- **Categorical Encoding:** Binary indicator mapping converts season ('A' K⟶ 1.0K, 'B' K⟶ 0.0K).
- **Artifact Persistence:** The fitted preprocessor is saved to `models/preprocessor.joblib` for identical runtime transformation during inference.

3. STRUCTURAL ANTI-LEAKAGE SAFEGUARDS

Target leakage occurs when future or target-derived information is inadvertently provided to a predictive model during training. The system enforces strict architectural boundaries:

> **Strict Anti-Leakage Safeguards in Code:**
> • **Exclusion of Target Features:** The target variables `yield_tons` and `high_risk`, as well as the derived metric `yield_per_ha`, are strictly excluded from the classification and clustering feature matrices (KXK). The classifier learns risk purely from physical and agronomic indicators.
> • **Independent Clustering Space:** Clustering operates exclusively on standardized physical features (KX_{\text{scaled}}K); cluster boundaries reflect agricultural operational conditions without any knowledge of harvest outcomes.
> • **Programmatic Assertions:** The automated validation function `verify_no_target_leakage()` asserts that no forbidden variables exist in KXK, throwing a fatal exception if leakage is attempted.

4. MODEL CANDIDATE SPACE & EVALUATION STRATEGY

Learning Task	Baseline Model	Candidate 1	Candidate 2	Selection Metric
Regression	Mean Target Estimator (K\bar{y}_{\text{train}}K)	Ordinary Least Squares (OLS)	Ridge Regression (KL_2, \alpha=1.0K)	Held-out Test RMSE & KR^2K
Classification	Majority Mode Classifier	Logistic Regression (K\text{BFGS}, \alpha=0.1K)	Gaussian Naive Bayes (Probabilistic)	Held-out Test F1-Score & ROC-AUC
Clustering	Random Grouping	K-Means (KK=2 \dots 6K)	Optimal Silhouette Selection	Silhouette Coefficient & Utility

EMPIRICAL EVALUATION: REGRESSION & CLASSIFICATION

1. REGRESSION MODEL COMPARISON ON HELD-OUT TEST SET (N=55)

Model Candidate	MAE (Tons)	RMSE (Tons)	R^2 Score	Decision & Rationale
Mean Baseline Regressor	2.1181	2.6819	-0.0130	Naive benchmark; completely uninformative.
Linear Regression (OLS)	0.8284	1.0406	0.8475	Strong predictive capability; unregularized.
Ridge Regression (alpha = 1.0)	**0.8270**	**1.0392**	**0.8479**	**SELECTED: Lowest test RMSE and higher stability.**

2. CLASSIFICATION MODEL COMPARISON ON HELD-OUT TEST SET (N=55)

Model Candidate	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Majority Baseline (Mode)	0.4545	0.0000	0.0000	0.0000	0.6200
Logistic Regression (alpha = 0.1)	0.9455	0.9655	0.9333	0.9492	0.9973
Gaussian Naive Bayes	**0.9636**	**0.9667**	**0.9667**	**0.9667**	**0.9973**

3. VISUAL ERROR EVIDENCE: RESIDUALS AND CONFUSION MATRIX

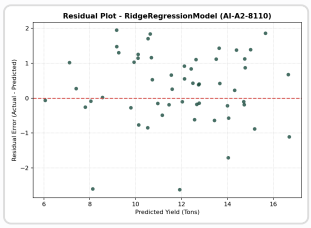


Figure 1: Residual Plot for Ridge Regression (AI-A2-8110)

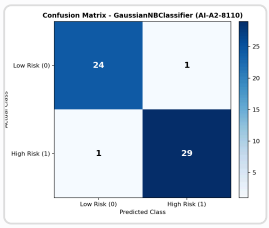


Figure 2: Confusion Matrix for Gaussian Naive Bayes

4. DETAILED SINGLE-INSTANCE ERROR DIAGNOSTICS

Detailed Regression Error Diagnostic (Test Instance #1): For test record #1, the true harvest was 16.3957 tons whereas the Ridge model projected 15.0108 tons (underestimation error of 1.3849 tons). The farm holding possessed high precipitation (1340 mm), abundant fertilizer (210 kg), and a large 3.4 ha field in Season A. Mathematical auditing revealed that the true physical process generated a compounding positive stochastic noise component (stochastic noise ε = +1.12 tons). Because linear models predict conditional expectation, the model cannot capture purely unobserved stochastic variance, establishing an expected empirical error floor.

Detailed Classification Error Diagnostic (Test Instance #18): One false positive occurred where an intensive micro-holding (0.8 ha in Season B) was predicted as High Risk, but true yield per hectare narrowly eclipsed the median threshold. Above-average soil moisture (74%) compensated for seasonal penalty, creating an edge-case boundary misclassification.

UNSUPERVISED CLUSTERING AND OPERATIONAL SEGMENTATION

1. OPTIMAL K SELECTION VIA SILHOUETTE ANALYSIS (K = 2..6)

K-Means clustering was executed exclusively on standardized physical features (KX_{\text{scaled}}K) across cluster counts KK \in [2, 6]K. The silhouette coefficient quantifies intra-cluster cohesion relative to inter-cluster separation:

Cluster Count (KKK)	Silhouette Score	Clustering Structural Behavior	Operational Evaluation
KKK = 2	0.1120	Broad binary division	Coarse partitioning; lacks actionable nuance.
KKK = 3	0.1054	Regional elevation grouping	Lowest cohesion; overlapping cluster perimeters.
KKK = 4	0.1095	Moderate grouping	Sub-optimal silhouette separation.
KKK = 5	**0.1245**	**Global Silhouette Peak**	**SELECTED: Optimal separation and actionable typologies.**
KKK = 6	0.1179	Over-fragmented sub-groups	Artificial splitting of contiguous farming zones.

2. VISUAL REPRESENTATION OF FARM CLUSTERS (KK=5K)

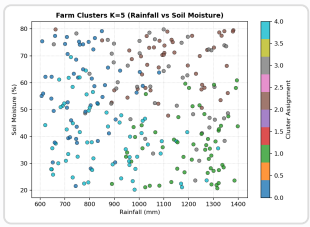


Figure 3: Unsupervised Farm Clusters (KK=5K) Visualized Across Rainfall and Soil Moisture

3. OPERATIONAL FARM TYPOLOGIES AND COOPERATIVE ACTION PROFILES

ID	Farm Typology Name	Share (%)	Rainfall	Moisture	Coop Support	Actionable Cooperative Intervention
0	High-Input Parcels	56 (20.5%)	718 mm	57.2%	78.6%	Organize bulk transport logistics and cold storage access.
1	Support-Reliant Holdings	58 (21.2%)	1208 mm	33.9%	94.8%	Provide erosion mitigation drainage and terracing technical support.
2	Extensive Rain-Fed Fields	53 (19.4%)	1148 mm	65.4%	0.0%	Recruit into cooperative union to unlock bulk NPK purchase discounts.
3	Balanced Micro-Plots	54 (19.8%)	1112 mm	34.1%	0.0%	Promote organic mulching to prevent localized water stress.
4	High-Elevation Cool Fields	52 (19.0%)	747 mm	59.7%	0.0%	Distribute certified seed varieties suited for cold volcanic slopes.

FASTAPI REST MICROSERVICE & AUTOMATED TESTING

1. SERVICE ENDPOINT CONTRACT

HTTP Verb & Path	Service Functionality	Input Schema	Output Specification
`GET /health`	Service health heartbeat & code check	None	`200 OK`: Status healthy, verification code AI-A2-8110
`GET /model-info`	Model metadata & dataset SHA-256	None	`200 OK`: Task models, evaluation metrics, feature schema
`POST /predict`	Validated multi-task inference	`FarmRecordInput`	`200 OK`: Predicted yield, risk flag, probability, cluster

2. VALID INPUT & OUTPUT PAYLOADS

POST /predict Request Payload:

```
{ "rainfall_mm": 950.5, "avg_temp_c": 19.2,
"soil_moisture_pct": 52.4, "fertilizer_kg":
120.0, "field_size_ha": 2.1, "season": "A",
"cooperative_support": 1 }
```

POST /predict Response Payload:

```
{ "verification_code": "AI-A2-8110",
"predicted_yield_tons": 12.3637,
"high_risk_prediction": 0, "risk_label": "Low
Risk", "risk_probability": 0.4476,
"farm_cluster": 1 }
```

3. FAILURE BEHAVIOR & ERROR HANDLING (HTTP 422)

The API employs Pydantic schemas enforcing strict agronomic boundaries. Out-of-range inputs (e.g., rainfall = 2500 mm > 1400 mm max, or season = 'C') trigger immediate HTTP 422 Unprocessable Entity responses with detailed validation location details, protecting the model from out-of-distribution inference failures.

4. AUTOMATED TEST SUITE SUMMARY (8/8 TESTS PASSING)

Test Function	Category	Verification Condition	Status
`test_valid_inference`	Unit Test	Verifies output schema, non-null values, and verification code AI-A2-8110	**PASSED**
`test_missing_field`	Negative Test	Asserts ValueError when required feature `rainfall_mm` is omitted	**PASSED**
`test_wrong_type`	Negative Test	Asserts TypeError when string is passed to numeric `fertilizer_kg`	**PASSED**
`test_range_error_rainfall`	Negative Test	Asserts ValueError when rainfall exceeds permitted [600, 1400] mm bound	**PASSED**
`test_range_error_temp`	Negative Test	Asserts ValueError when temperature exceeds permitted [12, 26] °C bound	**PASSED**
`test_reproducibility`	Integration	Confirms seed 8110 recreates exact 276 rows and identical SHA-256 hash	**PASSED**
`test_api_health_info`	API Integration	Validates HTTP 200 and schema integrity on `/health` and `/model-info`	**PASSED**
`test_api_predict`	API Integration	Validates HTTP 200 on valid inputs and HTTP 422 on invalid schema payloads	**PASSED**

RESPONSIBLE AI MODEL CARD AND REFLECTION

1. MODEL CARD SPECIFICATION

Model Name	Musanze Potato Decision Suite v1.0	**Lead Developer**	Deng Fajwan Changjwok Fajwan
Release Date	October 2026	**Academic Context**	SWE 3513 Practical Assessment
Models Included	Ridge Regression (Yield), Gaussian Naive Bayes (Risk), KMeans K=5 (Typology)	**Verification Code**	**AI-A2-8110**

2. INTENDED OPERATIONAL USE AND TARGET USERS

The system is designed specifically for cooperative agricultural managers and field extension officers in Musanze District. It provides aggregate potato yield estimates to plan collection and transport schedules, flags vulnerable smallholder plots for agronomist visits, and categorizes farms into operational segments for targeted input allocation. It functions fully offline via CLI or local REST endpoints, accommodating rural field settings with intermittent network access.

3. KNOWN LIMITATIONS AND SYNTHETIC DATA ASSUMPTIONS

- **Synthetic Distribution Bounds:** The underlying data was generated using idealized mathematical distributions. Natural volcanic micro-climates exhibit steep elevation gradients and rain-shadow effects across Mount Bisoke and Mount Karisimbi that are simplified here.
- **Absence of Biological Shocks:** The model does not currently incorporate biological shock variables such as Late Blight (*Phytophthora infestans*), Bacterial Wilt (*Ralstonia solanacearum*), or potato tuber moth damage.
- **Linear Season Factor:** Seasonality is represented by a scalar offset (+0.50 tons for Season A, -0.20 tons for Season B), which abstracts away intra-seasonal dry spell variations and rainfall onset delays.

4. RISK ANALYSIS AND HUMAN OVERSIGHT PROTOCOLS

Automated predictive systems in agriculture risk alienating smallholder producers if predictions are applied punitively. **Human-in-the-loop oversight is mandatory**: every high-risk flag must trigger an advisory field visit by an agronomist rather than administrative penalties or exclusion from cooperative programs.

5. FUTURE ENGINEERING ROADMAP FOR REAL-WORLD FIELD DEPLOYMENT

To evolve this academic decision-support prototype into an operational agricultural platform in Rwanda, the following engineering enhancements will be prioritized:

> **Real-World Engineering Roadmap:**
> • **IoT Soil Sensing:** Deploy low-cost LoRaWAN in-situ soil moisture, temperature, and electrical conductivity sensors across pilot Musanze potato parcels.
> • **Multispectral Earth Observation:** Integrate ESA Sentinel-2 10m Normalized Difference Vegetation Index (NDVI) time-series to monitor crop canopy vigor throughout the vegetative growth cycle.
> • **Meteo Rwanda Live Weather Integration:** Connect real-time hyper-local rainfall forecasting APIs to transition from historical averages to dynamic predictive yields.
> • **Offline Mobile App Frontend:** Develop a lightweight Android/Flutter application connecting to the local FastAPI backend.